AI.202

Robotic Perception

COURSE OVERVIEW

Fusing sensor data to understand the world: vision, LiDAR, depth, and multimodal perception for robots.



ILLUSTRATIVE HERO IMAGERY

ROBOTIC PERCEPTION

1

SESSION

60 MINUTES

ORIENTATION: PERCEIVING THE WORLD

KEY TOPICS

- Perception pipeline
- Uncertainty in sensing
- Modality trade-offs
- Perception in autonomy

TEACHING FOCUS

Give robots a model of their surroundings.

KEY TAKEAWAY Perception turns raw signals into world models.

ROBOTIC PERCEPTION

2

SESSION

60 MINUTES

VISUAL PERCEPTION

KEY TOPICS

- Camera models and calibration
- Stereo vision and depth
- Optical flow
- Object detection for robots

TEACHING FOCUS

Extract 3D structure from 2D images.

▲ **KEY TAKEAWAY** Stereo and flow reveal distance and motion.

3

SESSION

60 MINUTES

RANGE SENSING

KEY TOPICS

- LiDAR and point clouds
- Depth cameras
- Ultrasonic and infrared
- Point-cloud registration (ICP)

TEACHING FOCUS

Measure distance and shape directly.

KEY TAKEAWAY Range sensors give precise spatial answers.

4

SESSION

60 MINUTES

SENSOR FUSION

KEY TOPICS

- Why fuse sensors
- Kalman filter introduction
- Extended Kalman filter
- Complementary filtering

TEACHING FOCUS

Combine sensors for a better estimate.

KEY TAKEAWAY Fused sensors see more clearly than any alone.

ROBOTIC PERCEPTION

5

SESSION

60 MINUTES

PERCEPTION FOR NAVIGATION

KEY TOPICS

- Obstacle detection
- Free-space segmentation
- Place recognition
- Semantic mapping

TEACHING FOCUS

Support safe robot motion.

▲ **KEY TAKEAWAY** Perception answers 'where can I go?'.

ROBOTIC PERCEPTION

6

SESSION

60 MINUTES

PERCEPTION SYSTEMS & REVIEW

KEY TOPICS

- Real-time perception on robots
- Course recap and assessment
- Toward localization (AI.310)
- System integration

TEACHING FOCUS

Integrate perception into a robot loop.

▲ **KEY TAKEAWAY** Perception closes the loop with action.